

# Ventura County Environmental Justice Screen

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## Load Packages

```
library(tidyverse)
library(sf)
library(here)
library(tmap)
```

## Import Data

```
# Read in geodatabase of EJScreen data at the Census Block Group level

ejscreen <- sf::st_read(here::here("data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNMR

# Filter to state of California
california <- ejscreen |>
  dplyr::filter(ST_ABBREV == "CA")

# Filter to Ventura County
ventura <- ejscreen |>
  dplyr::filter(CNTY_NAME %in% c("Ventura County"))

# Read in city boundary data with city names for Ventura County

city_boundary <- sf::st_read(here::here("data", "City_Boundary", "City_Boundary.shp"))
```

## Visualize Data

```
# Switch mode to t_map viewing mode to interactive

tmap_mode("plot")

# Map 1

# Plot using Ventura County boundaries using ventura data and tm_polygons() filled with Percent Low Income

tm_shape(ventura) +
  tm_polygons(fill = "P_LOWINCPCT",

    # Incorporating color scale

    fill.scale = tm_scale(values = "YlOrBr"),

    # Adjust legend title

    fill.legend = tm_legend(title = "Percent Low Income")) +

  #Plotting only the city names from the city_boundary data and adjusting text size

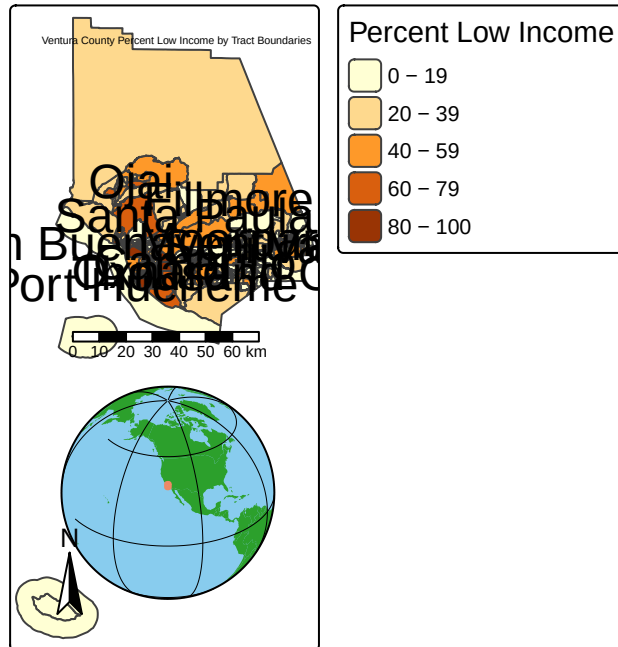
tm_shape(city_boundary) +
  tm_text("city_name", size = 1.5) +

  # Adding map title, scalebar, compass, and minimap

tm_layout(title = "Ventura County Percent Low Income by Tract Boundaries") +
tm_scalebar() +

  # Compass position to lower left

tm_compass(position = c("left", "bottom")) +
tm_minimap()
```



```
# Map 2

# Plot using Ventura County boundaries using ventura data and tm_polygons() filled with Air Toxics Cancer Risk

tm_shape(ventura) +
tm_polygons(fill = "CANCER",

            # Incorporating the same color scale

            fill.scale = tm_scale(values = "YlOrBr"),

            # Updating the legend title

            fill.legend = tm_legend(title = "Air Toxics Cancer Risk")) +

# Again, plotting only the city names from the city_boundary data and adjusting text size

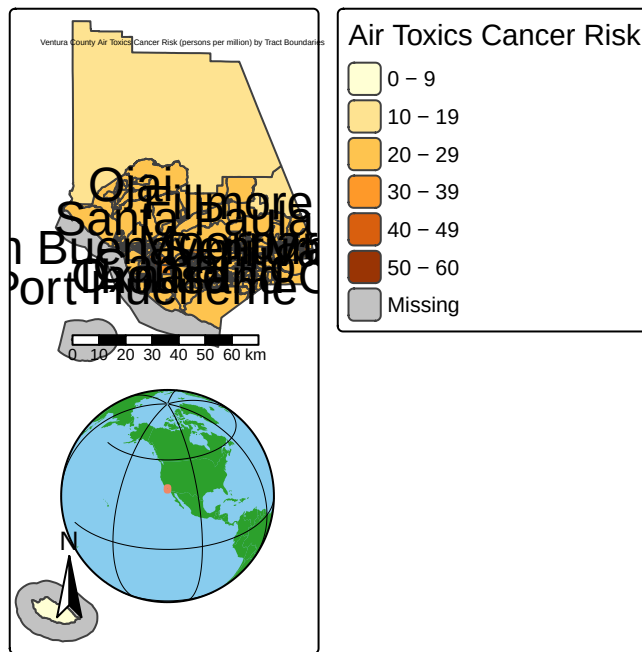
tm_shape(city_boundary) +
tm_text("city_name", size = 1.5) +

# Adding map title, scalebar, compass, and minimap
```

```
tm_layout(title = "Ventura County Air Toxics Cancer Risk (persons per million) by Tract Boundaries",
          tm_scalebar() +

          # Dropping compass position to lower left

          tm_compass(position = c("left", "bottom")) +
          tm_minimap())
```



## Map Interpretations

Below are two different maps conveying information about Ventura County which is located in Southern California between Santa Barbara County to the north and Los Angeles County in the south. Much like its northern and southern neighbors, Ventura County houses a diverse population of people and wealth disparities are very evident across city lines and within cities themselves. In the first map below, I chose to plot the Percent Low Income variable. Low income is defined as less than or equal to twice the poverty level which is a national number and the same across geographic regions. It is clear that many areas of Ventura County have a large percent of low income individuals (80-100%), particularly south of San Buenaventura and in large areas of Oxnard. I was interested if there was overlap of Air Toxic Cancer Risk (Lifetime cancer risk from inhalation of air toxics per million people) because of the presence of many agricultural fields along the 101 freeway in Oxnard and Camarillo which has adjacent

low-income areas. The second map displays that across Ventura County, the Air Toxics Cancer Risk is about 20-29 ppm. In areas of Camarillo and Oxnard, that value is higher at 30-39 ppm, and some areas within Oxnard have significantly higher values of 40-60 ppm. It is clear that in these low-income areas, the risk for developing cancer due to inhalation of air toxics is higher than in higher income areas.

## **Data Citations**

Data from the United States Environmental Protection Agency's previous EJ Screen: Environmental Justice Screening Tool

<https://drive.google.com/file/d/1nG6Nj1bXfzQFOVMO8Km3eNy4SWu1YcIQ/view>

Ventura County GIS Data

<https://venturacountydatadownloads-vcitsgis.hub.arcgis.com/datasets/city-boundary/explore>